

SARH, Chad

WMO: 647500

Lat: 9.150N

Lon: 18.383E

Elev: 365

StdP: 97.02

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	16.9	-4.4	2.7	32.5	-2.7	3.1	32.1	7.7	25.9	6.4	25.8	0.7	320	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.9	40.8	20.9	40.0	21.0	39.2	21.0	26.3	31.8	26.0	31.4	25.6	31.2	3.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.2	28.2	24.8	20.7	28.0	24.2	20.0	27.8	84.6	31.7	82.8	31.7	81.3	31.1	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.4	DB	13.1	42.4	1.9	0.5	11.7	42.7	10.6	43.0	9.6	43.3	8.2	43.6	(4)
(5)			WB	7.9	28.7	1.0	1.0	7.2	29.4	6.6	30.0	6.1	30.6	5.4	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	26.7	29.8	32.2	32.5	30.7	28.4	26.9	26.4	26.9	28.0	28.2	26.7	(6)
(7)		DBStd	2.73	2.20	2.10	1.58	1.91	2.30	1.89	1.66	1.35	1.36	1.30	1.34	1.59	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6790	517	554	687	676	643	552	524	509	507	557	544	519	(10)
(11)		CDD18.3	3748	258	321	428	426	385	302	266	251	257	299	294	261	(11)
(12)		CDH23.3	45067	3257	4459	6431	6366	5214	3375	2387	1971	2193	2954	3348	3112	(12)
(13)		CDH26.7	25307	1967	2836	4222	4075	2994	1566	913	667	855	1392	1938	1884	(13)
(14)	Wind	WSAvg	2.0	2.0	2.0	2.3	2.9	2.8	2.3	1.8	1.7	1.7	1.5	1.3	1.6	(14)
(15)	Precipitation	PrecAvg	1089	0	1	7	33	123	152	241	247	196	74	3	0	(15)
(16)		PrecMax	1743	0	17	50	114	460	306	647	641	518	199	31	5	(16)
(17)		PrecMin	764	0	0	0	0	22	30	71	83	87	0	0	0	(17)
(18)		PrecStd	222	0	4	12	33	90	62	111	104	93	48	7	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	41.2	41.9	41.6	40.2	37.2	34.9	33.7	34.2	36.2	37.9	38.1	(19)
(20)			MCWB	18.7	19.6	20.7	21.8	23.6	24.8	24.8	25.5	25.5	23.5	21.3	19.0	(20)
(21)		2%	DB	38.1	40.2	40.9	40.3	39.0	35.7	33.4	32.2	33.1	35.1	36.9	37.1	(21)
(22)			MCWB	18.4	19.1	20.3	22.8	23.9	24.8	25.0	25.0	25.5	24.0	21.0	19.2	(22)
(23)		5%	DB	37.0	39.2	40.1	39.5	37.9	34.4	32.2	31.2	32.2	34.1	36.0	36.1	(23)
(24)			MCWB	17.9	19.0	20.4	23.1	24.0	24.7	24.8	24.8	25.2	24.6	21.2	18.8	(24)
(25)		10%	DB	35.5	38.1	39.2	38.5	36.4	33.2	31.2	30.2	31.2	33.0	35.1	35.2	(25)
(26)			MCWB	17.2	18.7	20.2	23.2	23.9	24.6	24.6	24.7	24.9	24.7	21.5	18.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	23.5	25.3	25.8	26.6	26.6	26.2	26.4	26.7	26.7	25.9	24.4	(27)
(28)			MCDWB	34.1	36.5	35.9	35.3	34.0	32.9	31.1	30.4	31.1	31.9	33.0	33.6	(28)
(29)		2%	WB	20.0	22.2	24.3	25.1	25.7	25.9	25.7	25.7	26.1	26.2	25.1	22.5	(29)
(30)			MCDWB	34.2	34.9	35.8	35.1	33.2	31.9	30.6	30.0	30.9	31.4	32.1	30.5	(30)
(31)		5%	WB	18.8	21.4	23.6	24.7	25.2	25.5	25.3	25.4	25.7	25.7	24.4	21.0	(31)
(32)			MCDWB	33.9	34.2	35.3	34.8	33.0	31.5	30.3	29.6	30.3	30.9	31.3	30.9	(32)
(33)		10%	WB	17.8	20.3	22.9	24.2	24.8	25.2	25.0	25.0	25.4	25.3	23.7	19.7	(33)
(34)			MCDWB	32.9	34.0	34.6	34.4	32.6	31.1	29.9	29.1	29.8	30.6	30.4	30.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.5	17.3	14.8	11.9	10.6	9.4	8.2	8.0	9.0	10.5	15.2	18.1	(35)
(36)			MCDBR	19.2	18.0	16.0	13.1	12.2	10.9	9.9	9.4	10.3	12.1	16.2	18.5	(36)
(37)			MCWBR	6.5	6.5	5.5	3.8	3.4	3.6	3.6	3.8	4.2	3.9	4.3	5.6	(37)
(38)		5% WB	MCDWB	18.9	15.9	13.0	11.8	11.0	9.9	9.1	8.9	9.8	10.6	13.3	17.4	(38)
(39)			MCWBR	7.0	6.7	4.7	3.9	3.7	4.0	3.8	4.0	4.4	4.3	4.3	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.450	0.549	0.626	0.594	0.547	0.522	0.489	0.478	0.487	0.507	0.482	0.455	(40)	
(41)		taud	2.134	1.855	1.724	1.843	1.984	2.077	2.203	2.245	2.203	2.122	2.119	2.122	(41)	
(42)		Ebn at Noon	858	785	732	748	767	776	806	828	830	811	823	843	(42)	
(43)		Edh at Noon	155	211	244	214	182	163	145	142	149	160	156	154	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.09	6.26	6.34	6.22	5.88	5.38	4.95	4.80	5.27	5.73	6.00	5.87	(44)	
(45)		RadStd	0.19	0.23	0.34	0.29	0.21	0.22	0.22	0.27	0.22	0.23	0.19	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page